



CEPIITD

Online PG Diploma: Advanced Communication Engineering with Quantum and AI Integration

Course: Quantum Information and Security

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Indian Institute of Technology Delhi



Quantum Information and Security

Module-2

Quantum Information Theory

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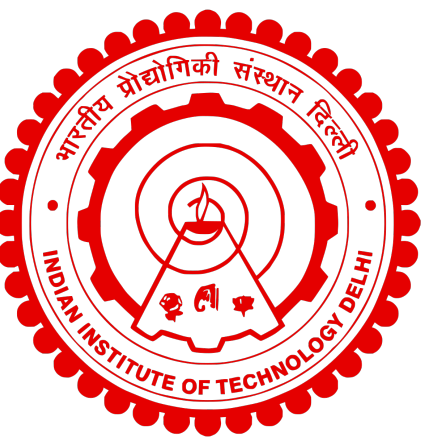
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Overview



- Quantum Information Concepts
- Quantum Entropy
- Quantum Data Compression
- Quantum Channels
- Quantum Capacity

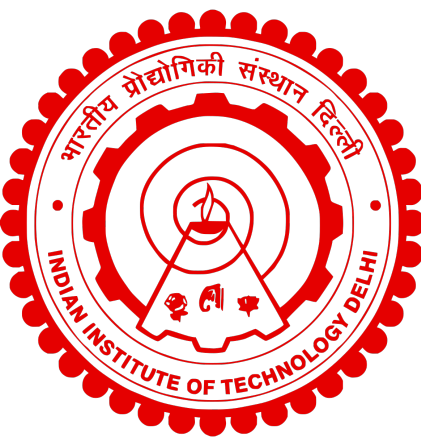
Schmidt Decomposition



- Every pure state $|\psi\rangle$ of a bipartite system (e.g., Alice + Bob) can be written in a special form called the **Schmidt decomposition**:

$$|\psi\rangle = \sum_{i=1}^d \lambda_i |i_A\rangle \otimes |i_B\rangle \text{ where:}$$

- $\lambda_i > 0$ are called **Schmidt coefficients**.
- $\{|i_A\rangle\}$ and $\{|i_B\rangle\}$ are orthonormal bases.
- $\sum_i \lambda_i^2 = 1$.
- $d \leq \min(\dim \mathcal{H}_A, \dim \mathcal{H}_B)$, where $\mathcal{H}_A$ and $\mathcal{H}_B$ are the Hilbert spaces of the two subsystems.
- The number of non-zero λ_i 's is called the **Schmidt rank**.



Schmidt Decomposition Example

- The Schmidt rank d tells us if a pure bipartite state is entangled.
 - If $d = 1$: the state is **not entangled** (product state).
 - If $d > 1$: the state is **entangled**.

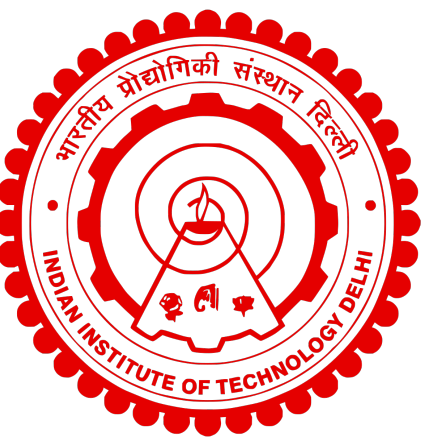
- Example: Consider the Bell state $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

- This can be written in Schmidt form:

$$|\psi\rangle = \lambda_1 |0_A\rangle |0_B\rangle + \lambda_2 |1_A\rangle |1_B\rangle \text{ with } \lambda_1 = \lambda_2 = \frac{1}{\sqrt{2}}.$$

- Since both coefficients are non-zero: $d = 2 \implies$ State is entangled

Partial Trace: Intuition



- In many situations, one part of the system (e.g., Bob's) is inaccessible.
- We then describe Alice's state using the **partial trace**: $\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|)$
- The result ρ_A is called the **reduced state** of Alice.
- The **partial trace** is the mathematical operation that "traces out" or "ignores" subsystem B
- This captures all the information Alice can obtain about her part of the system.

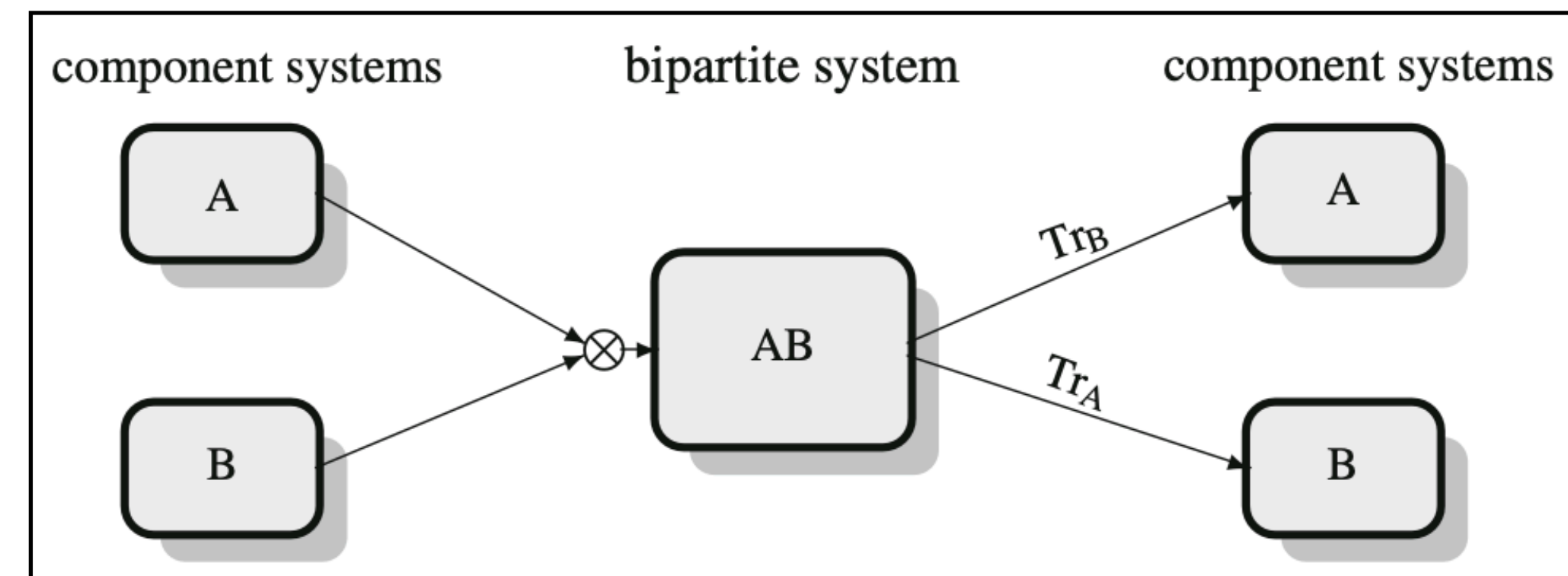
Partial Trace: Definition

- Consider a density operator ρ_{AB} acting on the combined Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$.
- Let $\{ |i_B\rangle \}_i$ be an orthonormal basis for subsystem B .
- The partial trace over B is defined as: $\text{Tr}_B(\rho_{AB}) = \sum_i (I_A \otimes \langle i_B |) \rho_{AB} (I_A \otimes |i_B\rangle)$.
- For simplicity, we usually write: $\rho_A = \text{Tr}_B(\rho_{AB}) = \sum_i \langle i_B | \rho_{AB} | i_B \rangle$.
- The state ρ_A is the reduced density matrix of subsystem A .
- **PT over B for elementary operators:** $\text{Tr}_B[|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|] = |a_1\rangle\langle a_2| \text{Tr}[|b_1\rangle\langle b_2|]$
where $\text{Tr}[|b_1\rangle\langle b_2|] = \langle b_1 | b_2 \rangle$.

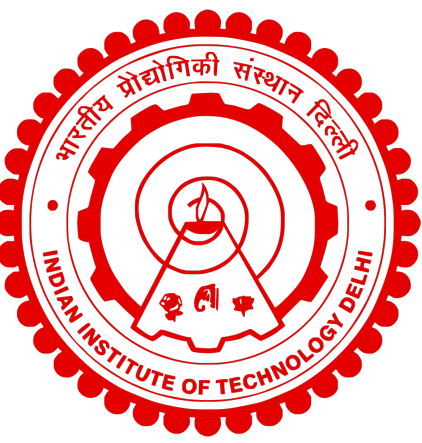
- **Proposition:** Let ρ be a density operator of the bipartite system $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ and let $\rho(a_1, a_2, b_1, b_2)$ be a matrix representation of ρ , where a_1, a_2 refers to a basis of $\mathcal{H}_A$ and b_1, b_2 to a basis of $\mathcal{H}_B$. Then the matrix representations of the reduced operators are given by

$$\rho_A(a_1, a_2) = \sum_b \rho(a_1, a_2, b, b), \text{ and}$$

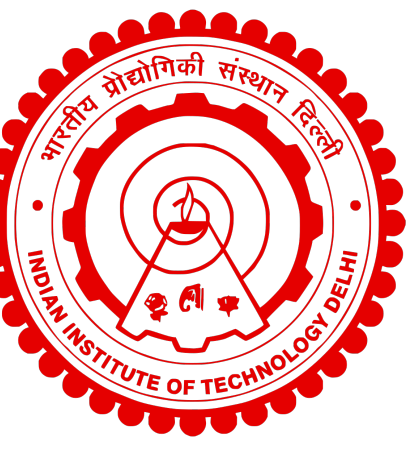
$$\rho_B(b_1, b_2) = \sum_a \rho(a, a, b_1, b_2).$$



Properties of Reduced State



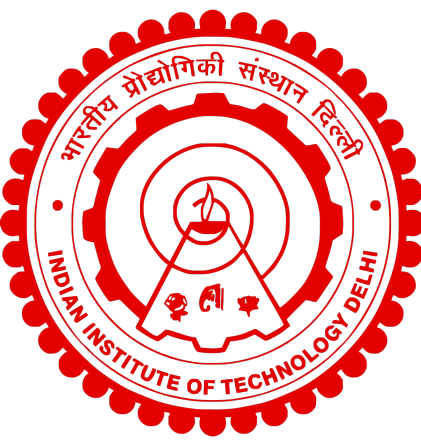
- ρ_A is a valid density operator:
 - It is **positive semi-definite**.
 - It has trace equal to 1.
- Taking partial traces in any order recovers the full trace: $\text{Tr}(\rho_{AB}) = \text{Tr}_A(\text{Tr}_B(\rho_{AB})) = \text{Tr}_B(\text{Tr}_A(\rho_{AB}))$.
- It describes the state of a subsystem when the rest of the system is ignored or inaccessible.
- Even if the overall state ρ_{AB} is **pure**, the reduced state ρ_A may be **mixed** due to entanglement.
- This tool helps analyze entanglement, decoherence, and measurement outcomes on subsystems.



Bell State Example

- Consider the Bell state: $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|0_A 0_B\rangle + |1_A 1_B\rangle)$.
- This is a pure entangled state of two qubits shared between Alice and Bob.
- Let's compute the reduced states $\rho_A = \text{Tr}_B(\rho_{AB})$ and $\rho_B = \text{Tr}_A(\rho_{AB})$.
- The full density operator is given by: $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$.
- Expand using the definition of $|\Phi^+\rangle$: $\rho_{AB} = \frac{1}{2} (|00\rangle + |11\rangle) (\langle 00| + \langle 11|)$.
- Multiply out the outer product: $\rho_{AB} = \frac{1}{2} (|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$.

Reduced State of A: $\rho_A = \text{Tr}_B(\rho_{AB})$



- We trace out Bob's system using the basis $\{ |0\rangle_B, |1\rangle_B \}$.

- Compute each term:

- $\text{Tr}_B(|00\rangle\langle 00|) = |0\rangle_A\langle 0| \cdot \underbrace{\langle 0|0\rangle}_1 = |0\rangle\langle 0|.$

- $\text{Tr}_B(|00\rangle\langle 11|) = |0\rangle_A\langle 1| \cdot \underbrace{\langle 0|1\rangle}_0 = 0.$

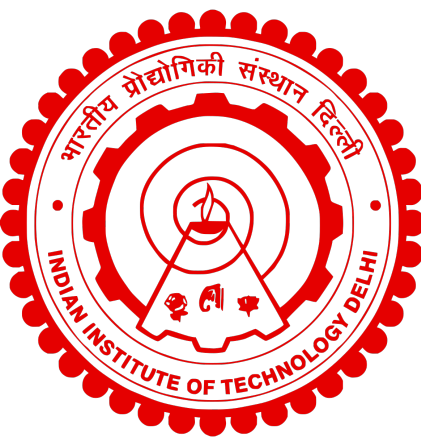
- $\text{Tr}_B(|11\rangle\langle 00|) = |1\rangle_A\langle 0| \cdot \underbrace{\langle 1|0\rangle}_0 = 0.$

- $\text{Tr}_B(|11\rangle\langle 11|) = |1\rangle_A\langle 1| \cdot \underbrace{\langle 1|1\rangle}_1 = |1\rangle\langle 1|.$

- So, $\rho_A = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I_{\mathcal{H}_A}.$

- This is the **maximally mixed state**.

Reduced State of B: $\rho_B = \text{Tr}_A(\rho_{AB})$



- Now trace out Alice's subsystem using basis $\{ |0\rangle_A, |1\rangle_A \}$.

- Compute each term:

- $\text{Tr}_A(|00\rangle\langle 00|) = |0\rangle_B\langle 0| \cdot \underbrace{\langle 0|0\rangle}_1 = |0\rangle\langle 0|.$

- $\text{Tr}_A(|00\rangle\langle 11|) = |0\rangle_B\langle 1| \cdot \underbrace{\langle 0|1\rangle}_0 = 0.$

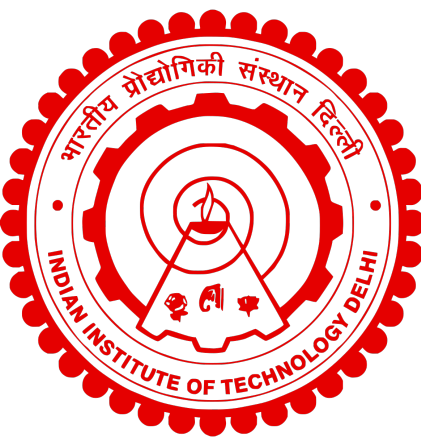
- $\text{Tr}_A(|11\rangle\langle 00|) = |1\rangle_B\langle 0| \cdot \underbrace{\langle 1|0\rangle}_0 = 0.$

- $\text{Tr}_A(|11\rangle\langle 11|) = |1\rangle_B\langle 1| \cdot \underbrace{\langle 1|1\rangle}_1 = |1\rangle\langle 1|.$

- So, $\rho_B = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I_{\mathcal{H}_B}$

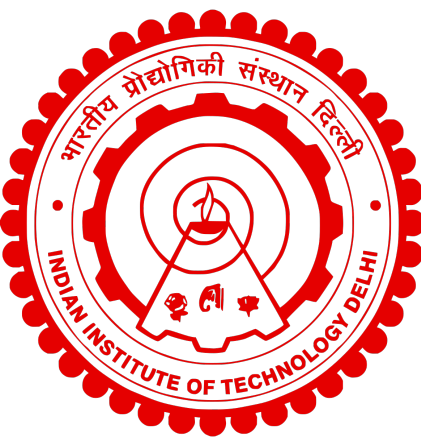
- Bob also sees a **maximally mixed state**.

What Does This Tell Us?



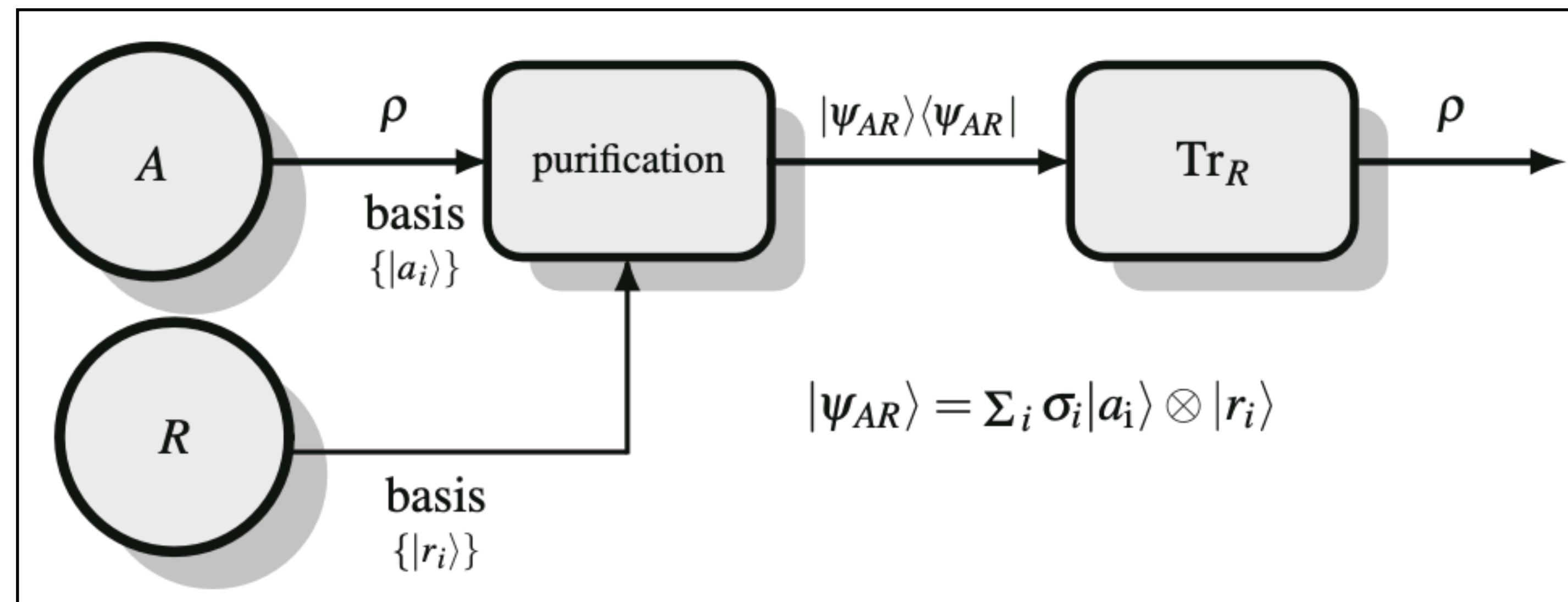
$$\text{But: } |\Phi^+\rangle_{AB}\langle\Phi^+| \neq \frac{1}{2}I_{\mathcal{H}_A} \otimes \frac{1}{2}I_{\mathcal{H}_B}!$$

- Both subsystems are in a maximally mixed state.
- But the total state ρ_{AB} is pure and entangled.
- The subsystems appear completely random if measured alone, but they are perfectly correlated when considered together.
- This is a signature of **quantum entanglement**.

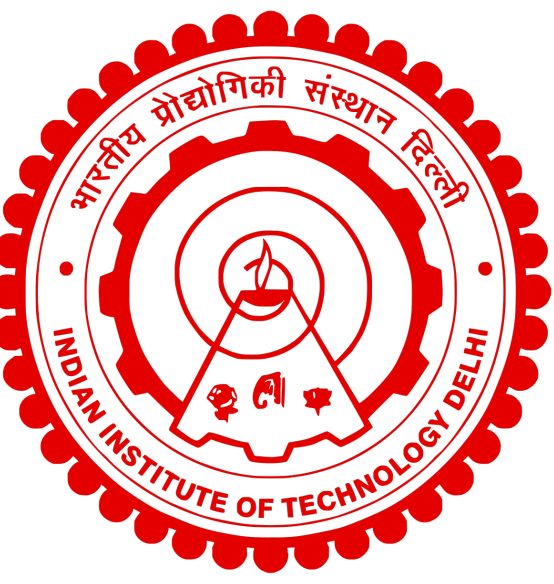


Purification of Mixed States

- **Purification:** A mixed state described by density operator can be represented as a pure state provided that the quantum environment is appropriately enlarged.
- For a quantum system A with density operator ρ_A , enlargement is obtained by introducing a copy R of the system A to form the joint system AR s.t. $\rho_A = \text{Tr}_R[|\psi_{AR}\rangle\langle\psi_{AR}|]$
- **Proposition:** Let ρ_A be a density operator of A and let $\rho_A = \sum_i \sigma_i^2 |a_i\rangle\langle a_i|$ be its eigenvalue decomposition. Then a **purification** of ρ_A is given by the composite state: $|\psi_{AR}\rangle = \sum_i \sigma_i |a_i\rangle \otimes |r_i\rangle$ where $\{r_i\}$ is a basis of R .

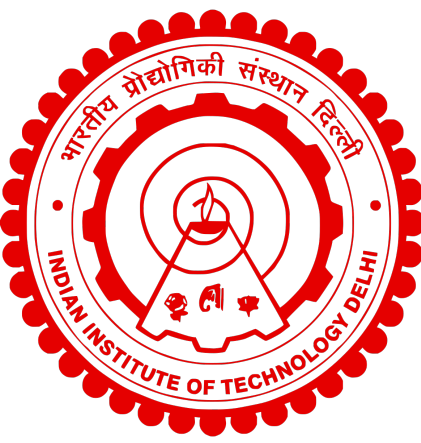


Overview



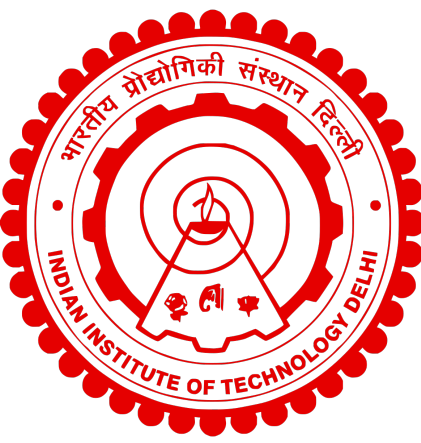
- Quantum Information Concepts
- **Quantum Entropy**
- Quantum Data Compression
- Quantum Channels
- Quantum Capacity

Motivation: Classical vs Quantum Entropy



- Classical entropy $H(X)$ measures uncertainty in a random variable based on its probability distribution.
- In the quantum setting, uncertainty arises both from classical randomness and quantum uncertainty.
- This is captured by the **density operator** ρ , which describes the quantum state.
- The quantum analogue of Shannon entropy is the **von Neumann entropy**.

Definition of von Neumann Entropy



- Let $\rho_A \in \mathcal{B}(\mathcal{H}_A)$ be a density operator of quantum system A .

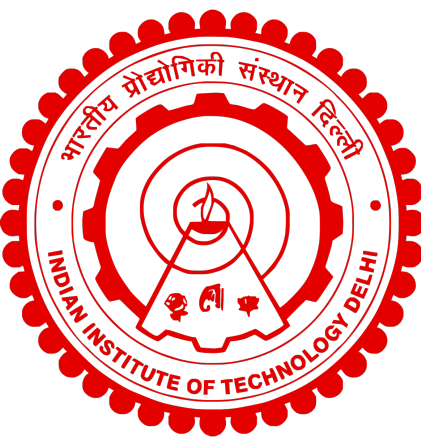
- **Von Neumann Entropy:** $S(A)_\rho = -\text{Tr}(\rho_A \log \rho_A)$

- **Notation:** $S(A)_\rho \equiv S(\rho_A) \equiv S(A)$

- If ρ_A has spectral decomposition $\rho_A = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j|$, then:

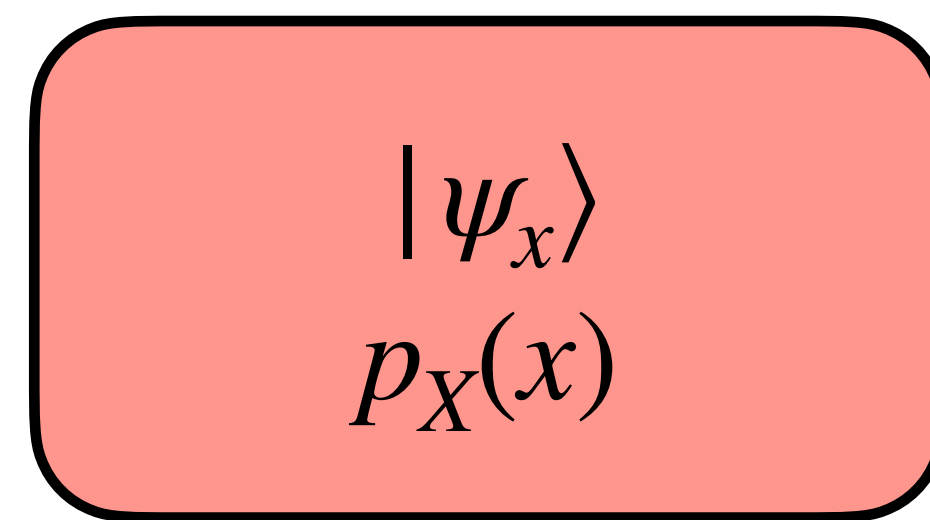
$$S(\rho_A) = - \sum_j \lambda_j \log \lambda_j$$

- The entropy is the function of eigenvalues λ_j of ρ .
- The eigenvalues form a probability distribution.
- Quantum entropy quantifies uncertainty in a quantum state, similar to classical entropy.

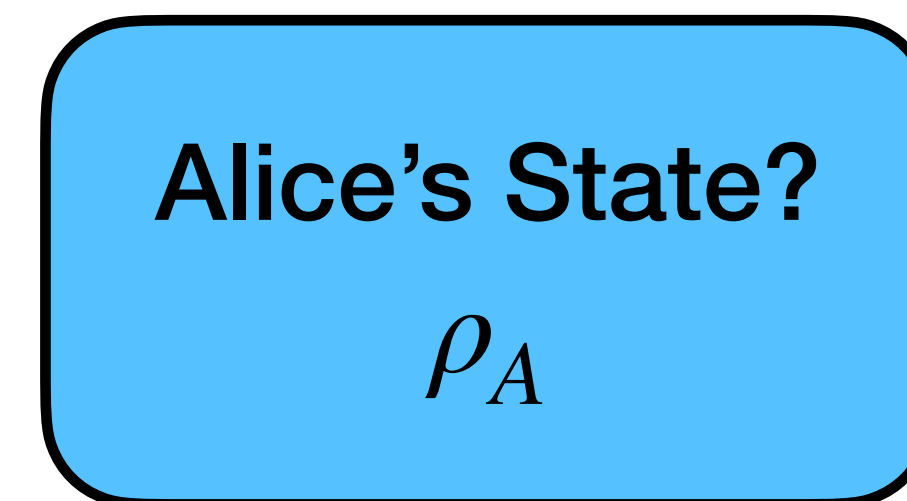


Interpretation and Examples

- **Example:** If Alice prepares a quantum state $|\psi_x\rangle$ with probability $p_X(x)$, the density matrix is:
$$\rho_A = \sum_x p_X(x) |\psi_x\rangle\langle\psi_x|$$
- The entropy $S(A)_\rho$ captures how much information Bob gains upon learning the state.

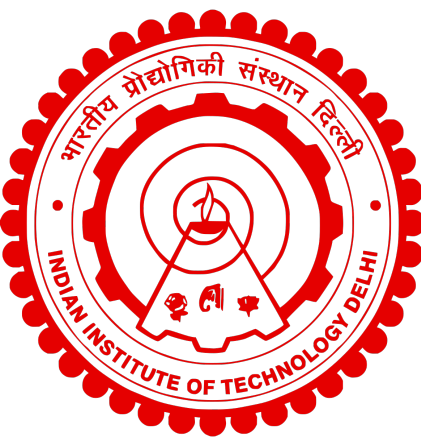


Alice

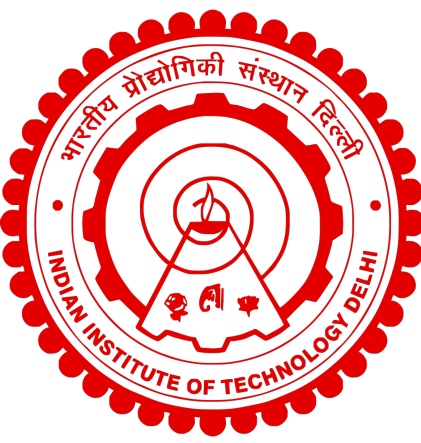


Bob

Properties of Quantum Entropy



- **Property-1 Entropy of Pure States:** The quantum entropy $S(\rho)$ vanishes iff ρ is a pure state.
 - If $S(\rho) = 0$, then $\lambda_j \log \lambda_j = 0 \ \forall j \implies \lambda_j = 0$ or $\lambda_j = 1$. Since $\sum_j \lambda_j = 1$, exactly one $\lambda_{j^*} = 1$.
 - If ρ is pure : $\rho = |\psi\rangle\langle\psi|$, then $S(\rho) = -\log(1) = 0$.
- **Property-2 Non-negativity of Entropy:** The quantum entropy $S(\rho)$ is always non-negative $S(\rho) \geq 0$
 - This follows from the fact that the Shannon entropy is non-negative and $\{\lambda_j\}$ form a valid probability distribution.
- **Property-3 Maximum Entropy:** The quantum entropy is bounded above by $\log(d)$, where d is the dimension of the Hilbert space: $S(\rho) \leq \log d$
 - The maximum is achieved for the maximally mixed state: $\rho = \frac{1}{d}I$.
 - Its eigenvalues are uniform: $\lambda_j = \frac{1}{d}$ for all j .
 - Then $S(\rho) = -\sum_j \frac{1}{d} \log \frac{1}{d} = \log d$.



Properties of Quantum Entropy

- **Property-4 Invariance Under Isometries:** Entropy is invariant under isometries:

- $S(\rho) = S(V\rho V^\dagger)$

$$V\rho V^\dagger = \sum_j \lambda_j V |\psi_j\rangle \langle \psi_j| V^\dagger$$

Reason:

- $= \sum_j \lambda_j |\psi'_j\rangle \langle \psi'_j|$

- The eigenvalues remain unchanged, hence entropy remains the same.

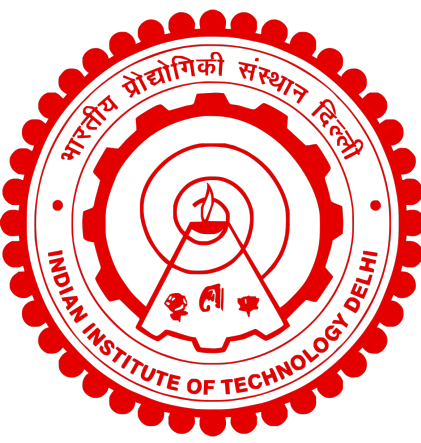
- **Property-5 Concavity of Quantum Entropy:** Quantum entropy is concave in the density operator.

- Let $\{p_X(x), \rho_x\}$ be a set of density operators and probabilities, then

$$S\left(\sum_x p_X(x) \rho_x\right) \geq \sum_x p_X(x) S(\rho_x)$$

- **Intuition:** Mixing quantum states increases uncertainty.

Quantum Entropies



- These properties enable quantum generalizations of classical quantities like conditional entropy, joint entropy, and mutual information.

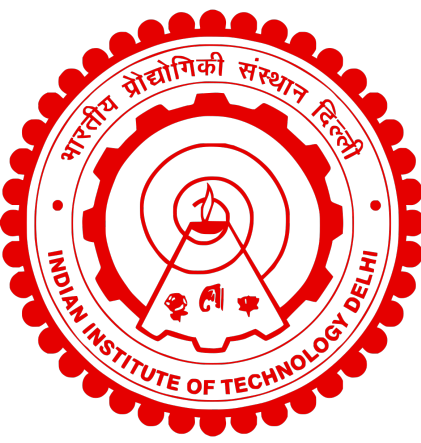
**Joint
Entropy**

**Conditional
Entropy**

**Coherent
Information**

**Mutual
Information**

Joint Quantum Entropy

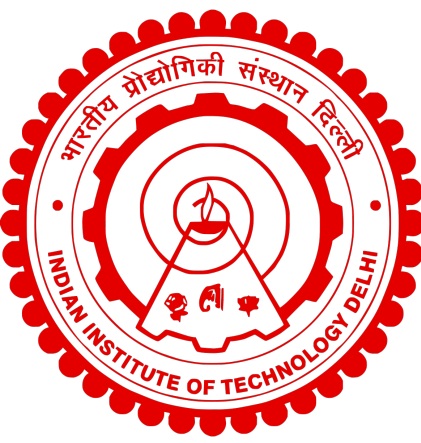


- **Definition:** Let $\rho_{AB} \in \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B)$ be the density operator of a bipartite system. The **joint entropy** is defined as:

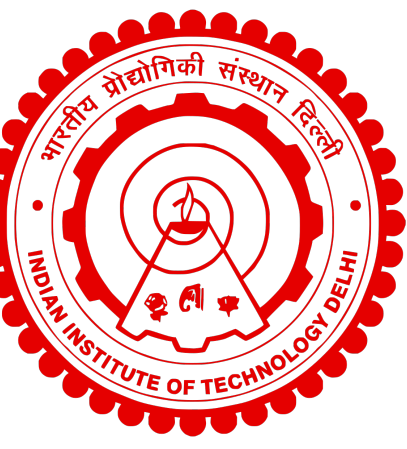
$$S(AB)_\rho = -\text{Tr}(\rho_{AB} \log \rho_{AB})$$

- This is a natural extension of the quantum entropy to composite systems.
- For a multipartite state ρ_{ABC} , we define $S(AB)_{\rho_{ABC}} = S(\text{Tr}_C(\rho_{ABC}))$.
- **Properties:**
 1. Additivity of quantum entropy under tensor product: $S(A, B) = S(A) + S(B)$ if $\rho_{AB} = \rho_A \otimes \rho_B$
 2. Sub-additivity inequality: $S(A, B) \leq S(A) + S(B)$
 3. Triangle inequality: $S(A, B) \geq |S(A) - S(B)|$

Quantum vs Classical Joint Entropy



- **Classical case:** $H(X, Y) \geq H(X)$, $H(X, Y) \geq H(Y)$
- **Quantum case:** Not always!
 - Consider a pure bipartite state: $|\psi\rangle_{AB} = \sum_i \lambda_i |e_i\rangle_A |f_i\rangle_B$
 - Marginal states: $\rho_A = \sum_i \lambda_i^2 |e_i\rangle_A \langle e_i|$, $\rho_B = \sum_i \lambda_i^2 |f_i\rangle_B \langle f_i|$
 - Then, $S(AB)_\psi = 0$, $S(A)_\psi = S(\rho_A) \neq 0$, $S(B)_\psi = S(\rho_B) \neq 0$



Marginals of Pure States

- **Theorem:** Let $|\psi\rangle_{AB}$ be a pure bipartite state. Then:

$$S(A)_\psi = S(B)_\psi, \quad S(AB)_\psi = 0$$

- **Additivity for Product States:** $S(\rho_A \otimes \sigma_B) = S(\rho_A) + S(\sigma_B)$

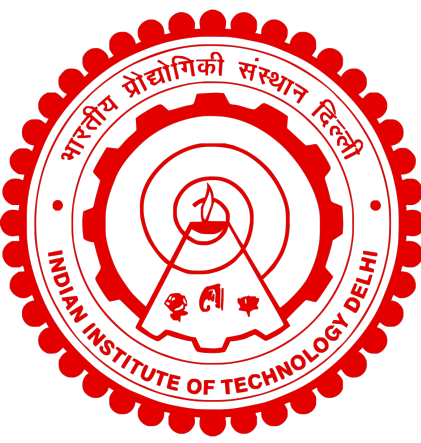
- **Example:** Consider hybrid classical-quantum state: $\rho_{AZ} = \sum_z p_Z(z) \rho_z^A \otimes |z\rangle_Z \langle z|$

- Then: $S(AZ)_\rho = S(Z) + \sum_z p_Z(z) S(\rho_z^A)$

- **Example:** Bipartite Entangled State, Bell state: $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|0_A 0_B\rangle + |1_A 1_B\rangle)$.

- Marginal states: $\rho_A = \frac{1}{2}I, \quad \rho_B = \frac{1}{2}I$

- Entropies: $S(AB)_{\Phi^+} = 0, \quad S(A)_{\Phi^+} = S(B)_{\Phi^+} = \log 2 = 1$



Conditional Entropy

- **Definition:** The conditional entropy of a bipartite state ρ_{AB} is:

$$S(A | B) = S(AB) - S(B)$$

- **Classically:**

- Entropies are always non-negative
- Hence $S(A | B) \geq 0$
- No need to define coherent information

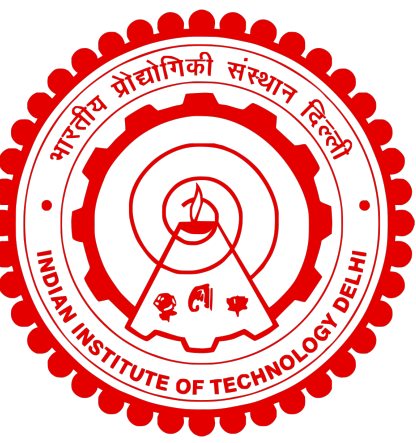
- **Quantumly:**

- $S(A | B)$ can be negative!
- Motivates the need for **coherent information**.

- **Example:** Bell State $|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|0_A 0_B\rangle + |1_A 1_B\rangle)$.

- **Pure state** $|\Phi^+\rangle \Rightarrow S(AB) = 0$ and $S(A) = S(B) = 1$
- **Conditional Entropy:** $S(A | B) = S(AB) - S(B) = 0 - 1 = -1$

Entanglement and Conditional Entropy



- **Theorem:** A pure bipartite quantum state $|\psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ is entangled if and only if:

$$S(A|B)_\psi < 0$$

- Since $|\psi\rangle_{AB}$ is pure: $S(AB)_\psi = 0 \Rightarrow S(A|B)_\psi = -S(B)_\psi$
- So equivalent statement: $|\psi\rangle_{AB}$ is entangled $\iff S(B)_\psi > 0$
- **Proof:** $\Rightarrow |\psi\rangle_{AB}$ entangled $\Rightarrow$ Schmidt rank $d > 1$

- Bob's marginal state: $\rho_B = \sum_{i=1}^d \lambda_i^2 |i\rangle_B \langle i|$

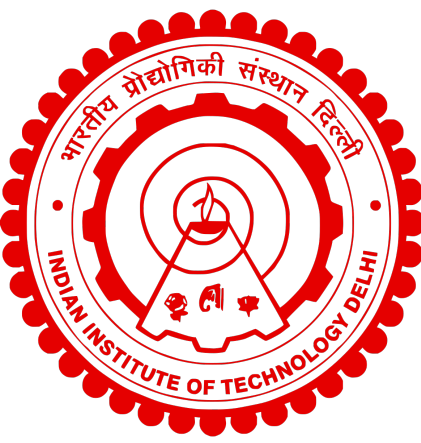
- Then: ρ_B not pure and $S(\rho_B) > 0$

- **Proof:** $\Leftarrow$ Assume $S(B)_\psi > 0$, then $S(\rho_B) = -\sum_{i=1}^d \lambda_i^2 \log(\lambda_i^2)$

- If $d = 1$, then $\lambda^2 = 1 \Rightarrow S(B) = 0$

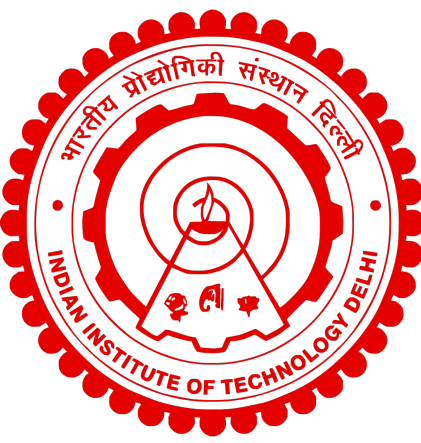
- So $S(B)_\psi > 0 \Rightarrow d > 1 \Rightarrow |\psi\rangle_{AB}$ is entangled

Why Coherent Information?



- **Issue:** Classical conditional entropy $H(A | B)$ can't be negative, but in quantum information theory it can.
- **Solution:** Define coherent information: $I(A \rangle B)_\rho = S(B)_\rho - S(AB)_\rho = -S(A | B)_\rho$
- **Interpretation:**
 - When $S(A | B) < 0$, $I(A \rangle B) > 0$
 - Coherent information quantifies how much quantum information is preserved through a noisy channel

Quantum Mutual Information

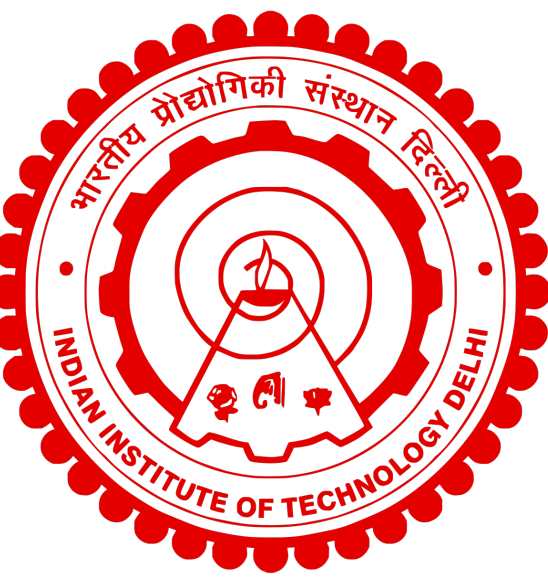


- **Definition:** For a bipartite quantum state ρ_{AB} , the mutual information is:

$$\begin{aligned} I(A : B)_\rho &= S(A)_\rho + S(B)_\rho - S(AB)_\rho \\ &= S(A)_\rho - S(A | B)_\rho \\ &= S(B)_\rho - S(B | A)_\rho \end{aligned}$$

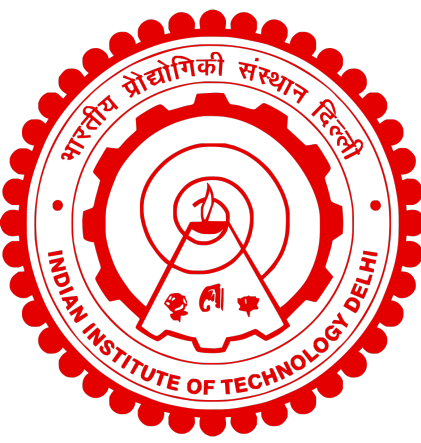
- **Interpretation:** Measures total correlations (classical + quantum) between A and B .
- **Non-Negativity of Mutual Information:** $I(A : B) \geq 0$
- **Proof:** Conditioning does not increase entropy:
 $S(A | B) \leq S(A) \Rightarrow S(A) - S(A | B) \geq 0 \Rightarrow I(A : B) \geq 0$
- **Example:** Consider the Bell state: $|\Psi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$
 - **Properties:** Pure state: $S(AB) = 0$
 - **Marginals:** $\rho_A = \rho_B = \frac{1}{2}I \Rightarrow S(A) = S(B) = 1$
 - So, Mutual Information: $I(A : B) = 1 + 1 - 0 = 2$

Overview



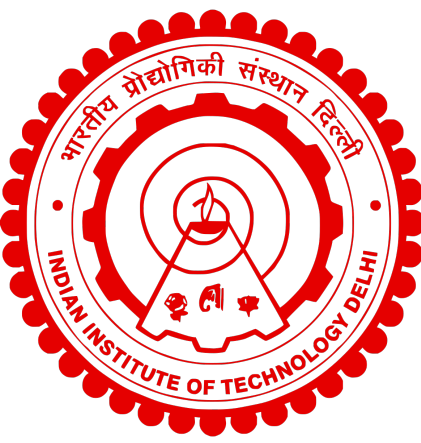
- Motivation
- Quantum Entropy
- **Quantum Data Compression**
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Bit vs Qubit in Information Theory



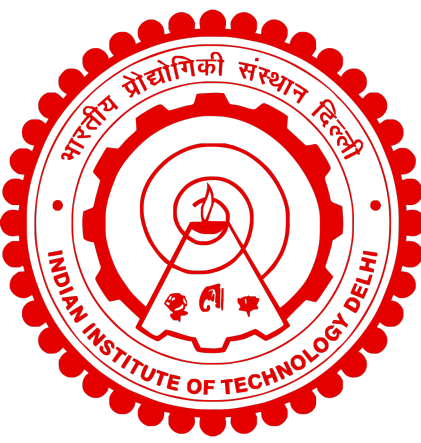
- In Quantum IT, the **logarithm of the dimensionality** of the space is the measure of the information content of a system and plays the role of the logarithm of the size of a codebook in Classical IT
- CIT: a codebook $\mathcal{C} = \mathcal{A}^L$ with $|\mathcal{A}| = d$ contains d^L codewords and $\log |\mathcal{C}| = L \log d$, so that each codeword is represented by $L \log d$ bits.
- In QIT: we consider L -long sequences of states of the form $|\Psi\rangle = |\psi_{i_1}\rangle \otimes \dots \otimes |\psi_{i_L}\rangle$ where $|\psi_{i_j}\rangle \in \mathcal{H}$, so that $|\Psi\rangle \in \mathcal{H}^{\otimes L}$. If $\dim \mathcal{H} = d$, then $\log \dim \mathcal{H}^{\otimes L} = L \log d$.
- If we call *qudit* the basic Hilbert space $\mathcal{H}$, then the sequence $|\Psi\rangle$ is represented by L qudits, or equivalently by 1 qudit/symbol. For qubits: $\mathcal{H} = \mathcal{Q}$, with $\dim \mathcal{Q} = 2$

Source Coding Theorems



- **Question:** What is the minimum number of bit per symbol or qubit per symbol needed to represent a codeword or a state sequence? Answer: Source Coding Theorems
- **Shannon's Source Coding Theorem:** For classical i.i.d symbols with probability distribution $\{p_i\}$, as $L \rightarrow \infty$, a minimum of $H(A)$ bits/symbol are required for encoding
- **Schumacher's Source coding theorem:** In quantum case the probability distribution is encoded in density operator as $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ and determines the quantum entropy of a state $S(\rho)$, which becomes $LS(\rho)$ for the L -long sequence. In the limit the minimum number is given by a quantum entropy $S(\rho)$, expressed in qubit per symbol.

Quantum Data Compression



- **Aim:** Find an encoding protocol that maps the states of a Hilbert space into a new Hilbert space of reduced dimensionality, with the possibility of recovering the original states.
- Let $d = \dim \mathcal{H}$ and its L -extension $\mathcal{H}^{\otimes L}$ of dimension d^L where the source state sequences (quantum sourcewords) are defined.
- Map these sequences into quantum codewords in a new Hilbert space $\mathcal{H}_c$ s.t.
 $d_c := \dim \mathcal{H}_c < \dim \mathcal{H}^{\otimes L} = d^L$



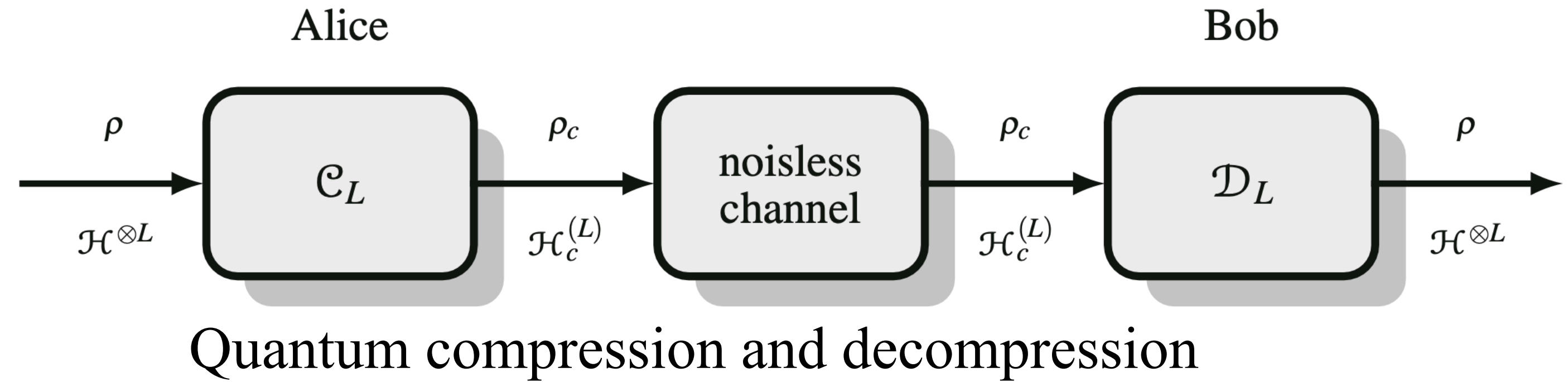
Schumacher's Compression Protocol

- Quantum compression and decompression:

$$\mathcal{C}_L : \mathcal{H}^{\otimes L} \rightarrow \mathcal{H}_c^{(L)}, \quad \mathcal{D}_L : \mathcal{H}_c^{(L)} \rightarrow \mathcal{H}^{\otimes L}$$

- Compression Rate:

$$R := \frac{\log \dim \mathcal{H}_c^{(L)}}{\log \dim \mathcal{H}^{\otimes L}} = \frac{\log \dim \mathcal{H}_c^{(L)}}{L \log d}$$



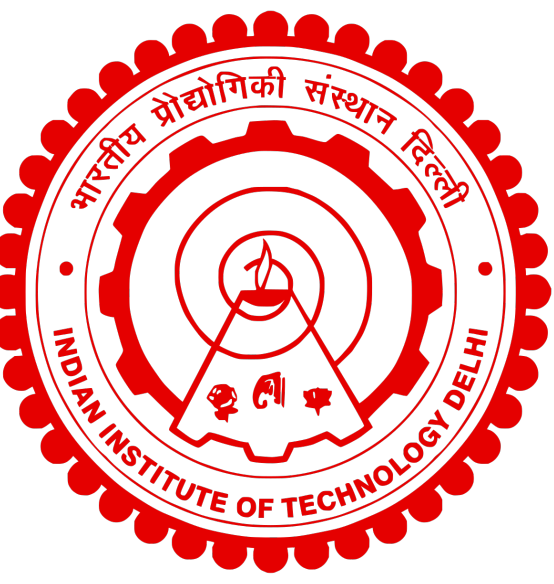
- If $\mathcal{H}^{\otimes L}$ consists of L qubits ($d = 2$), then $R = \log \dim \mathcal{H}_c^{(L)} / L$, and $\dim \mathcal{H}_c^{(L)} = 2^{LR}$, thus R reads in qubits/symbol.
- $\mathcal{C}_L$ compresses ρ of $L \log d$ qubits into state ρ_c of LR qubits and $\mathcal{D}_L$ decompresses ρ_c into ρ .

- Fidelity:** $F(\rho, \sigma) = \left(\text{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2$

s.t. $0 \leq F \leq 1$ and becomes $F = 1$ only when the two states coincide.

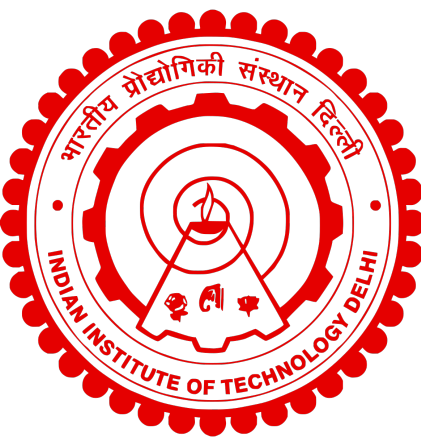
- Theorem:** (Schumacher's noiseless quantum channel coding) *Consider an IID quantum source with entropy $S(\rho)$ and rate R . Then, if $R > S(\rho)$ there exists an asymptotically reliable compression protocol (where $F \rightarrow 1$ as $L \rightarrow \infty$). Conversely, if $R < S(\rho)$ any compression protocol is not reliable.*

Overview



- Motivation
- Quantum Entropy
- Quantum Data Compression
- **Quantum Channels**
- Quantum Capacity

Quantum Channels



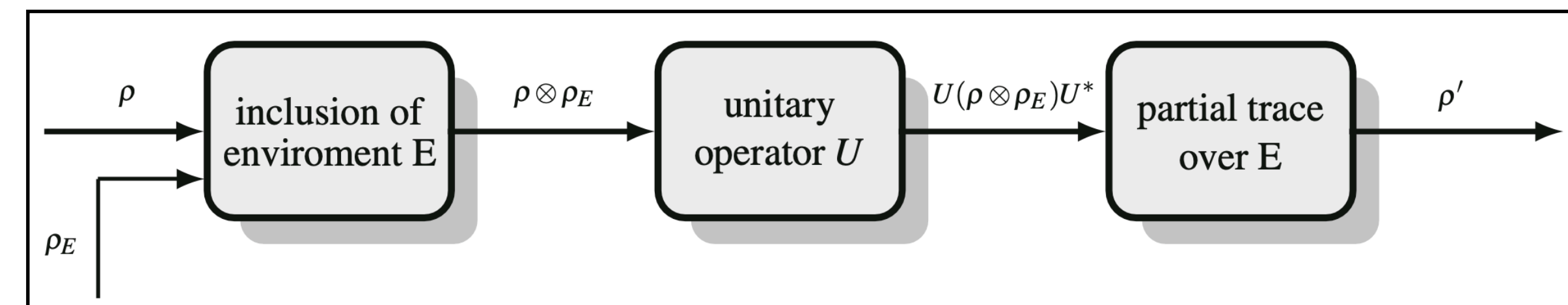
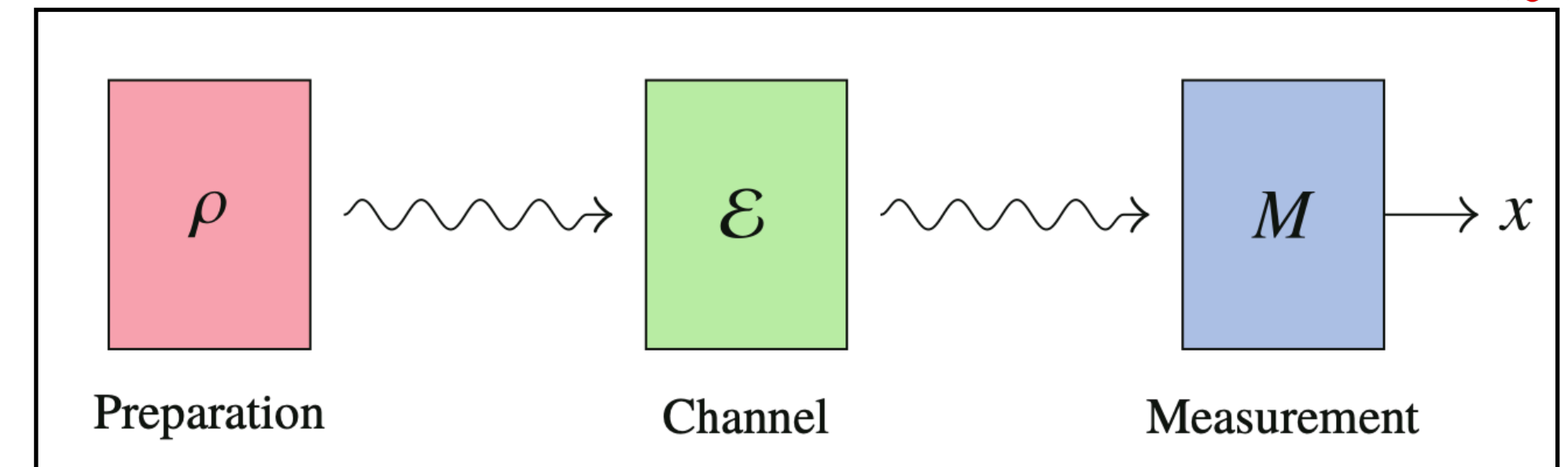
- Recall: Schematic of a quantum experiment
- Quantum channels** describe the dynamics of quantum systems, mapping states from a Hilbert space $\mathcal{H}_A$ to $\mathcal{H}_B$.
- Mathematically:** A quantum channel is an operator:

$$\mathcal{E} : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$$

where $\mathcal{B}(\mathcal{H})$ denotes the set of bounded operators on the Hilbert space $\mathcal{H}$.

- Composite space** $\mathcal{H} \otimes \mathcal{H}_E$: Add an ancillary Hilbert space $\mathcal{H}_E$ (environment state ρ_E) to the input space $\mathcal{H}$
- Unitary evolution of composite state with output state (using partial trace):

$$\rho' = \mathcal{E}[\rho] = \text{Tr}_E [U (\rho \otimes \rho_E) U^*]$$



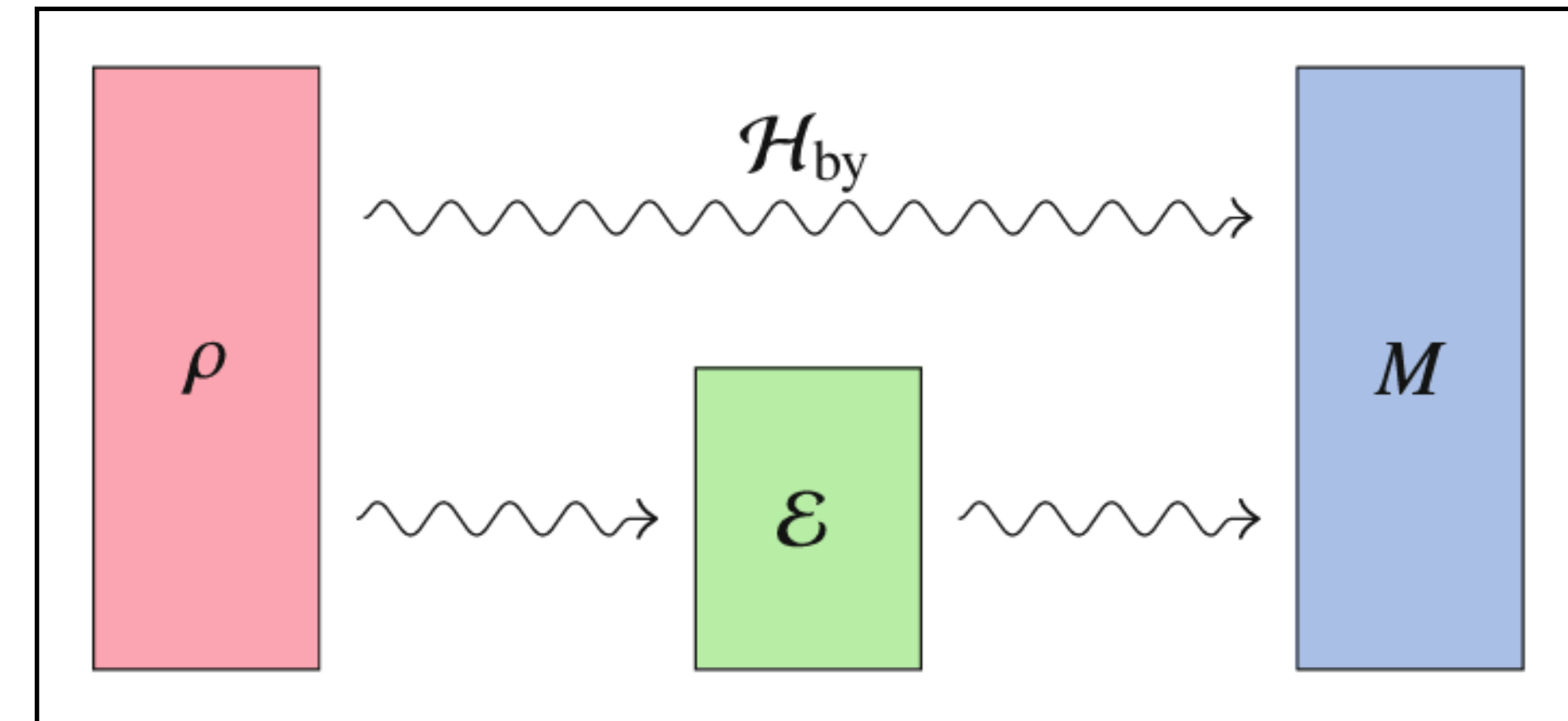
Unitary representation of a noisy quantum channel

Quantum Channels

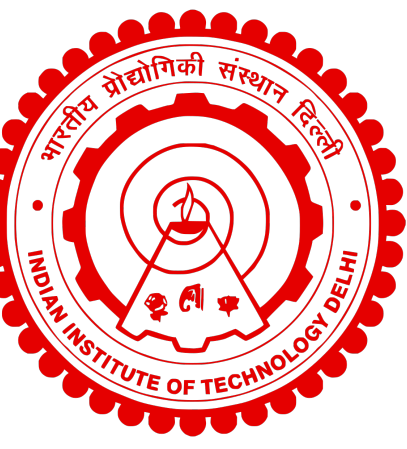
- **Definition:** A quantum channel is a map that satisfies the following properties:

1. **Linearity:** $\mathcal{E}(\alpha\rho + \beta\sigma) = \alpha\mathcal{E}(\rho) + \beta\mathcal{E}(\sigma)$ for all $\rho, \sigma \in \mathcal{B}(\mathcal{H}_A)$ and $\alpha, \beta \in \mathbb{C}$.
2. **Complete Positivity:** $\mathcal{E}$ is completely positive if for all $n \in \mathbb{N}$, the extended map $\mathcal{E} \otimes \text{id}_n : \mathcal{B}(\mathcal{H}_A \otimes \mathbb{C}^n) \rightarrow \mathcal{B}(\mathcal{H}_B \otimes \mathbb{C}^n)$ maps positive operators to positive operators. This is necessary to ensure positivity when part of a larger entangled system is acted on.
3. **Trace Preservation:** Ensures output is a valid density matrix:

$$\text{Tr}(\mathcal{E}(\rho)) = \text{Tr}(\rho) \quad \text{for all } \rho \in \mathcal{B}(\mathcal{H}_A)$$



- **Summary:** A quantum channel is a linear, completely positive, and trace-preserving map $\mathcal{E} : \mathcal{B}(\mathcal{H}_A) \rightarrow \mathcal{B}(\mathcal{H}_B)$ satisfying the three above properties.
- These conditions ensure that $\mathcal{E}$ maps valid quantum states to valid quantum states in all contexts.



Kraus Representation

- **Theorem (Choi-Kraus):** A map $\mathcal{E}$ is a quantum channel if and only if there exists a set of operators $\{K_j\}_{j=1}^d$ such that

$$\mathcal{E}(\rho) = \sum_{j=1}^d K_j \rho K_j^\dagger, \text{ where}$$

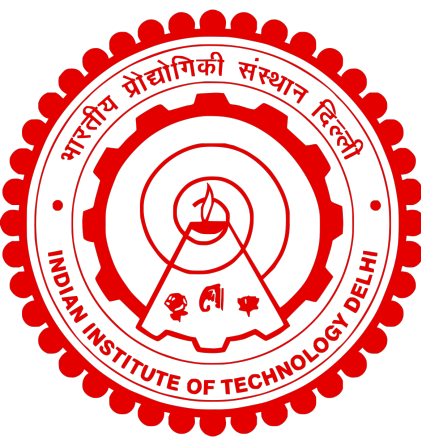
$$K_j : \mathcal{H}_A \rightarrow \mathcal{H}_B \quad \forall j \in \{1, \dots, d\} \quad \text{with } d \leq \dim(\mathcal{H}_A) \dim(\mathcal{H}_B) \text{ and}$$

$$\sum_{j=1}^d K_j^\dagger K_j = I_A.$$

- **Examples:**

- **Unitary Channel:** $\mathcal{E}(\rho) = U\rho U^\dagger \equiv \mathcal{U}(\rho)$ where U is unitary.
 - Only one Kraus operator. It is easy to reverse such a channel.
 - **Reversible:** $(\mathcal{U}^\dagger \circ \mathcal{U})(\rho) = U^\dagger U \rho U^\dagger U = \rho$
- **Isometric Channel:** $\mathcal{E}(\rho) = V\rho V^\dagger$ with $V^\dagger V = I$, $V : \mathcal{H}_A \rightarrow \mathcal{H}_B$.
- **Note:** Kraus decomposition is not unique; different sets of $\{K_j\}$ can describe the same channel.

Quantum Channels Examples



- **Amplitude Damping Channel:** It models energy dissipation (e.g., spontaneous emission) and describes the decay from the excited state $|1\rangle$ to the ground state $|0\rangle$ with probability γ , where $0 \leq \gamma \leq 1$.

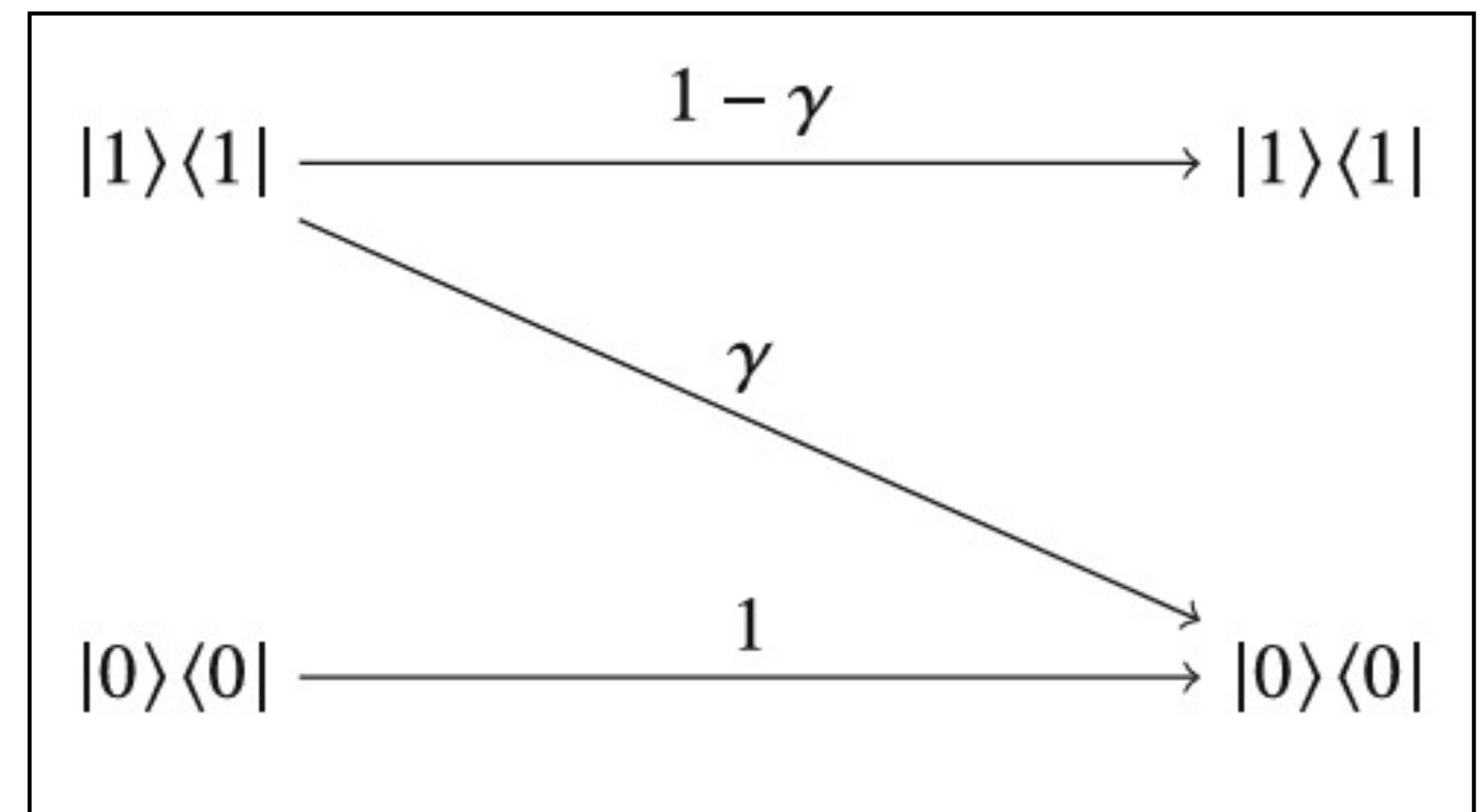
$$K_1 = \sqrt{\gamma} |0\rangle\langle 1|, \quad (\text{decay from } |1\rangle \rightarrow |0\rangle)$$

- Kraus Representation:

$$K_2 = |0\rangle\langle 0| + \sqrt{1-\gamma} |1\rangle\langle 1|, \quad (\text{no decay})$$

- These satisfy the completeness relation: $K_1^\dagger K_1 + K_2^\dagger K_2 = I$

- **Interpretation:** With probability γ , the system decays to $|0\rangle$, and with probability $1 - \gamma$, it remains in its state.



Quantum Channels Examples

- **Bit flip channel:**

$$\mathcal{E}(\rho) = p\sigma_x\rho\sigma_x + (1-p)\rho.$$

- Kraus representation:

$$d_E = 2, K_1 = \sqrt{1-p}I_2, K_2 = \sqrt{p}\sigma_x$$

- **Depolarising channel:**

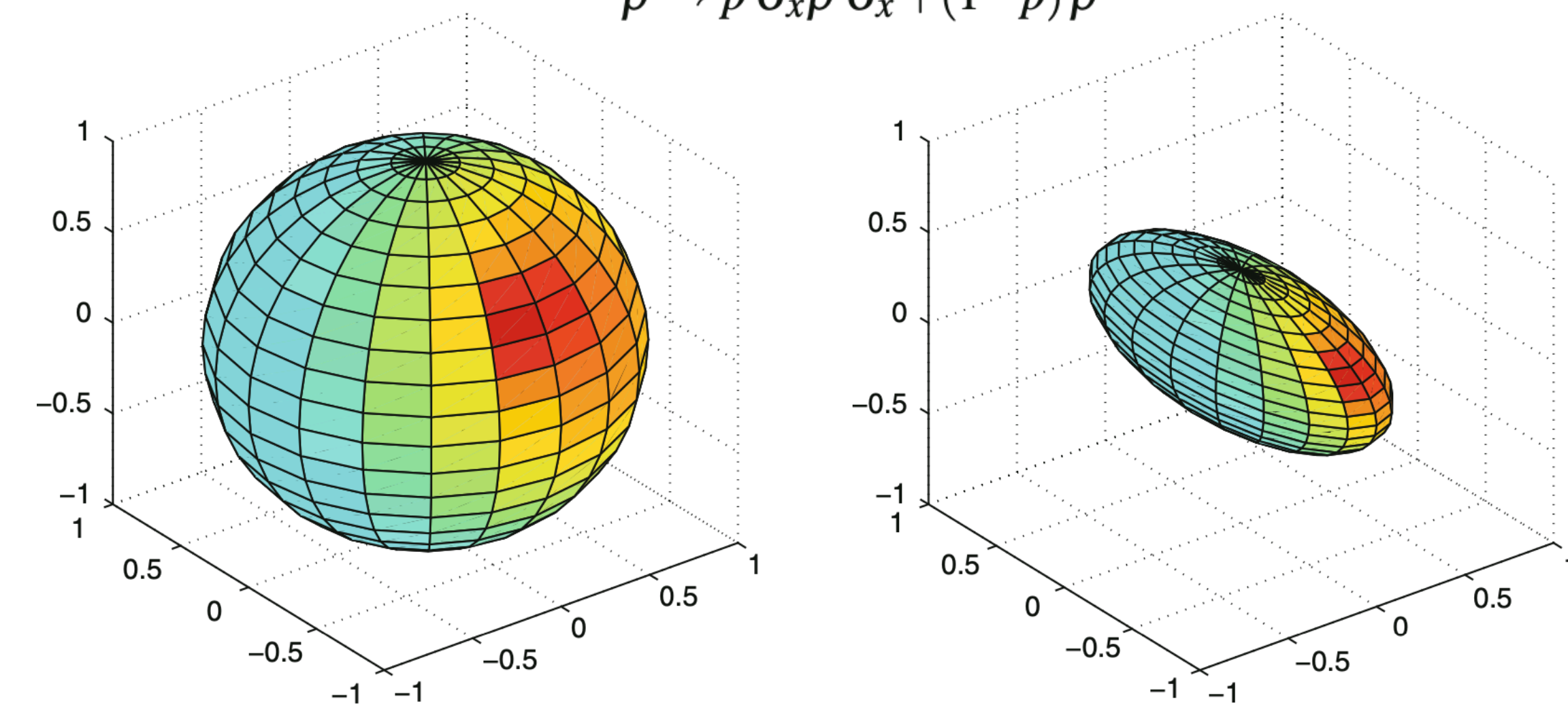
$$\mathcal{E}(\rho) = (1-p)\rho + p(1/d)I_{\mathcal{H}}.$$

- Kraus representation can be obtained for finite-dimensional Hilber space.

- For qubit space: $\mathcal{H} = \mathcal{Q} = \mathbb{C}^2$:

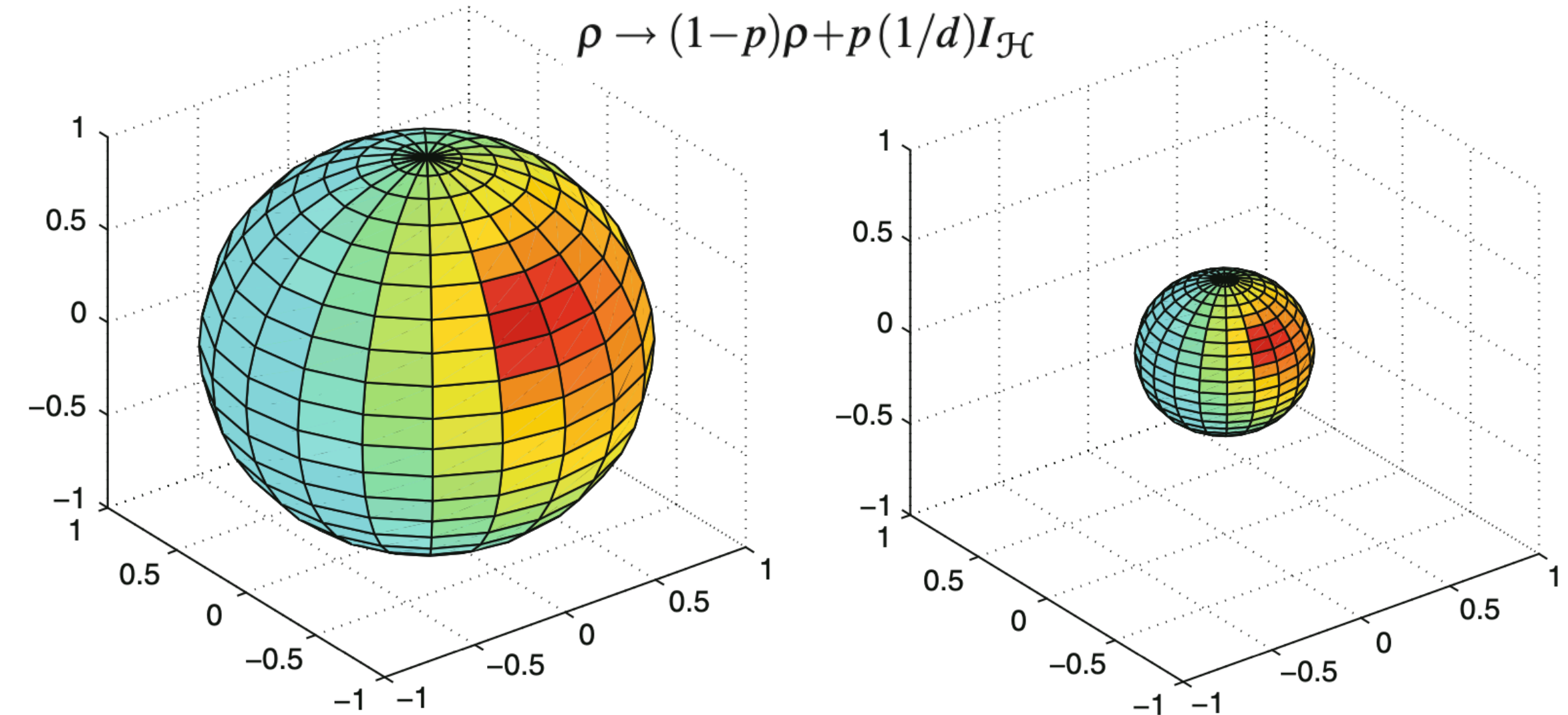
$$I_{\mathcal{H}} = I_2 = \frac{1}{2} \sum_{i=0,x,y,z} \sigma_i \rho \sigma_i^*.$$

$$\rho \rightarrow p\sigma_x\rho\sigma_x + (1-p)\rho$$



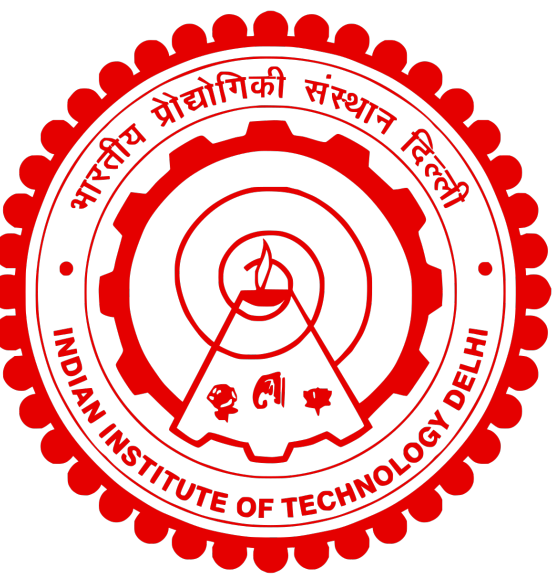
Interpretation of the **bit-flip channel** with the *Bloch sphere*.

$$\rho \rightarrow (1-p)\rho + p(1/d)I_{\mathcal{H}}$$



Interpretation of the **Depolarizing Channel** with the *Bloch sphere*.

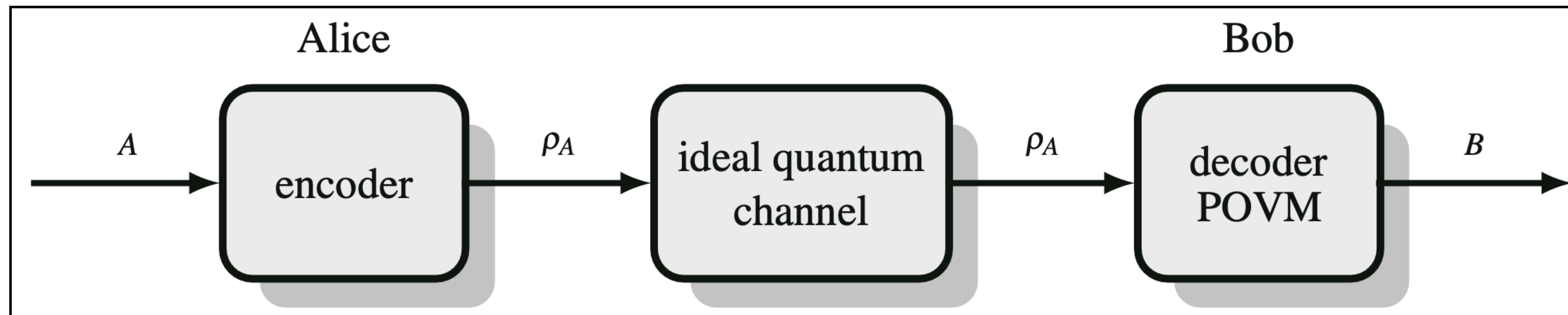
Overview



- Motivation
- Quantum Entropy
- Quantum Data Compression
- Quantum Channels
- **Quantum Capacity**

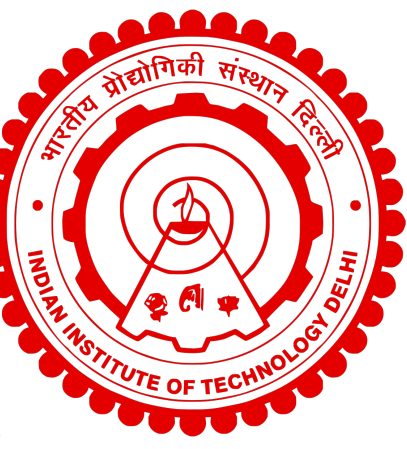
Accessible Information

- **Classical-Quantum Communications:** Classical information transfer Alice $\rightarrow$ Bob with classical source at Tx.: $A \simeq (\mathcal{A}, p_A)$ and receiver side sees a classical source: $B \simeq (\mathcal{B}, p_{A|B})$



- **Mutual Information:** $I(A; B) = H(A) - H(A | B)$
quantifies how much information A and B have in common
- **Accessible Information:** The maximum mutual information obtained through optimum POVM measurement

$$I_{\text{acc}} := \max_{\text{POVM}} I(A; B)$$



Holevo Information

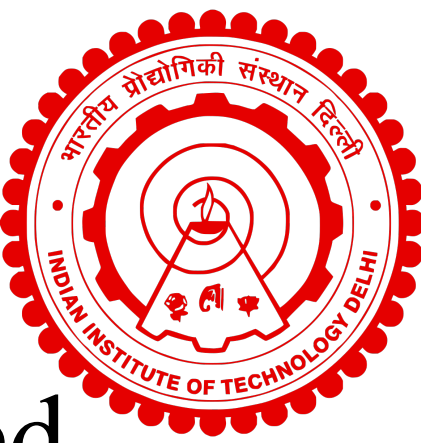
- **Holevo Entropy:** Ensemble of K quantum states $\mathcal{L} = \{(p_1, \rho_1), \dots, (p_K, \rho_K)\} = \{p_a, \rho_a\}$ then χ -information or Holevo- χ is defined as:

$$\chi(\{p_a, \rho_a\}) := S(\rho) - \sum_a p_a S(\rho_a)$$

$$\text{where } \rho = E[\rho_A] = \sum_a p_a \rho_a \text{ (Avg. Density operator)}$$

- **Holevo- χ properties:**

1. **Entropy of the constellation:** Holevo- χ depends only on the ensemble $\mathcal{L}$ i.e., $\chi(\mathcal{L})$
2. **Non-negativity:** $\chi(\mathcal{L}) \geq 0$ (due to concavity of quantum entropy).
3. **For pure states:** $S(\rho_a) = 0$, thus $\chi(\{p_a, \rho_a\}) = S(\rho)$
4. A quantum operation Φ can never increase the Holevo- χ , i.e., $\mathcal{L}' = \{(p_a, \Phi(\rho_a))\}$ then $\chi(\mathcal{L}') \leq \chi(\mathcal{L})$



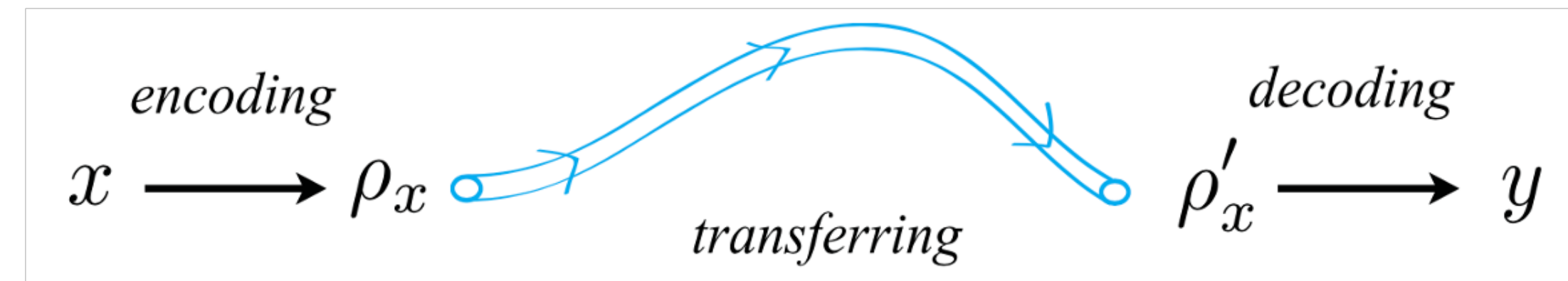
Holevo Bound

- **Theorem (Holevo's bound):** For any measurement Bob can do, the mutual information is limited by the Holevo- χ :

$$I(A; B) \leq \chi(\{p_a, \rho_a\})$$

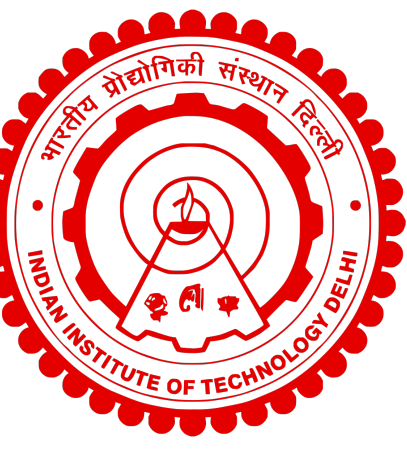
- The conditions for the equality in Holevo Bound are:
 1. The density operators ρ_a commute (are simultaneously diagonalizable) and
 2. Bob perform the **POVM** measurement with the common eigenbasis of the $\{\rho_a\}$
- **Classical Quantum Communication:**
- **HSW Theorem:** Classical capacity of quantum channels

$$C(\Phi) = \max_{p_x, \rho_x} \left\{ S \left(\sum_x p_x \mathcal{E}[\rho_x] \right) - \sum_x p_x S(\mathcal{E}[\rho_x]) \right\}$$



Quantum Channel : $\mathcal{E}[\cdot]$

$$\rho'_x = \mathcal{E}[\rho_x]$$



Example: Holevo Bound

- **Example:** A qubit system where Alice prepares $|\psi_0\rangle = |0\rangle$, $|\psi_1\rangle = \cos \theta |0\rangle + \sin \theta |1\rangle$ with prob. $p_0 = p$ and $p_1 = 1 - p$.

- Density operators: $\rho_0 = |\psi_0\rangle \langle \psi_0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\rho_1 = |\psi_1\rangle \langle \psi_1| = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$

- Avg. density operator: $\rho = p\rho_0 + (1 - p)\rho_1 = \begin{bmatrix} (1 - p)\cos^2 \theta + p & (1 - p)\cos \theta \sin \theta \\ (1 - p)\cos \theta \sin \theta & (1 - p)\sin^2 \theta \end{bmatrix}$

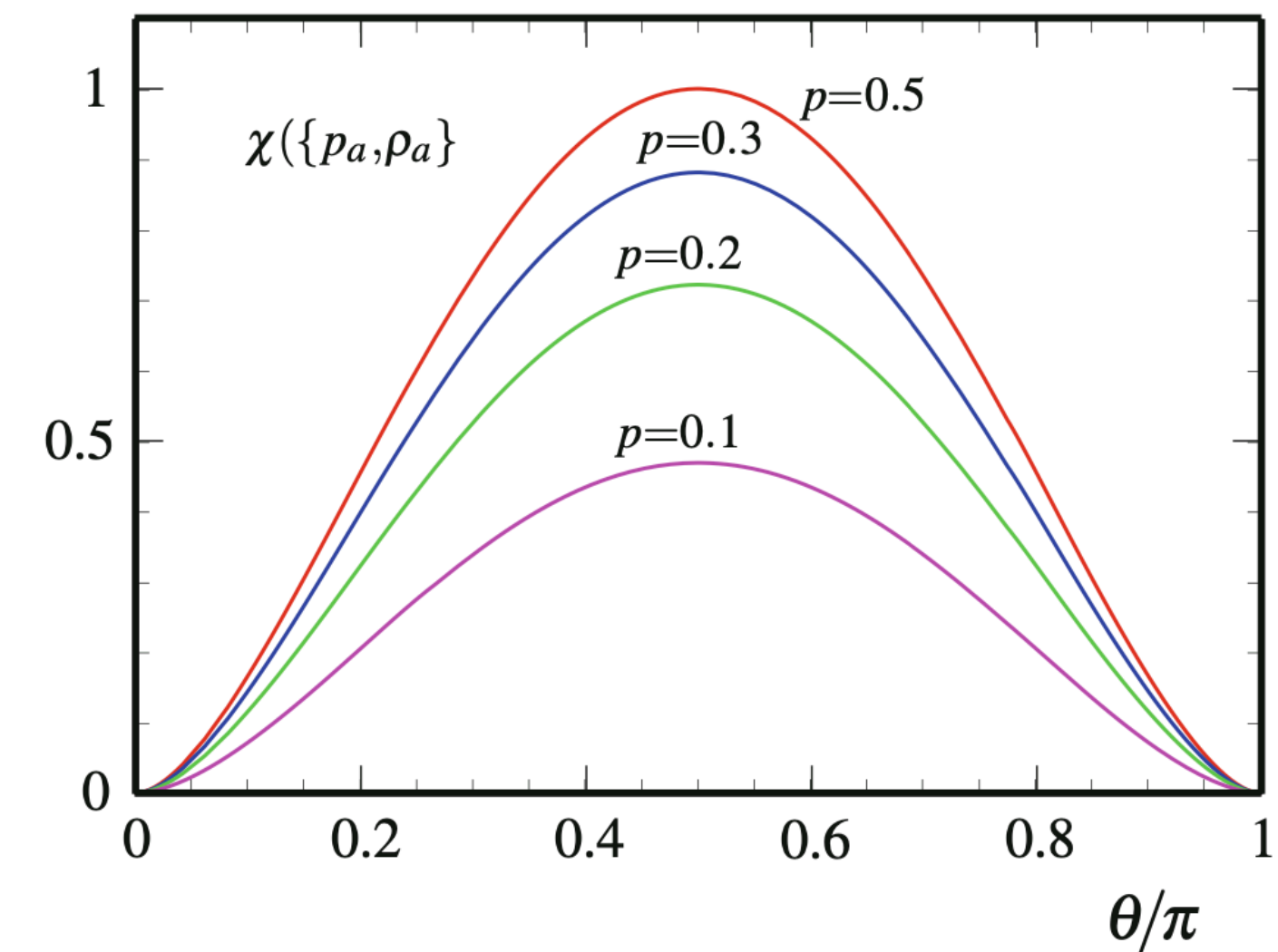
- Note for pure states: $S(\rho_0) = S(\rho_1) = 0$

- Eigenvalues of ρ :

$$\lambda_{\pm} = \frac{1}{2} \left[1 \pm \sqrt{2p^2 - 2(p - 1)\cos 2\theta p - 2p + 1} \right]$$

then:

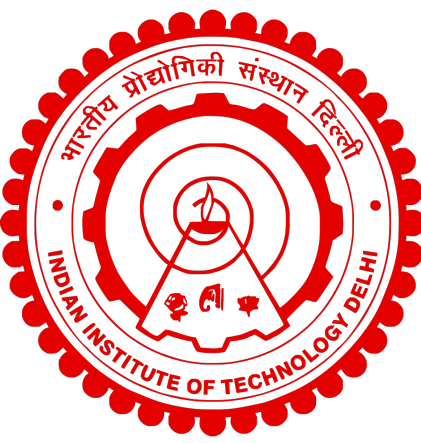
$$\chi(p_a, \rho_a) = S(\rho) = H(\{\lambda_-, \lambda_+\})$$



Holevo bound as a function of the parameter θ in a qubit system

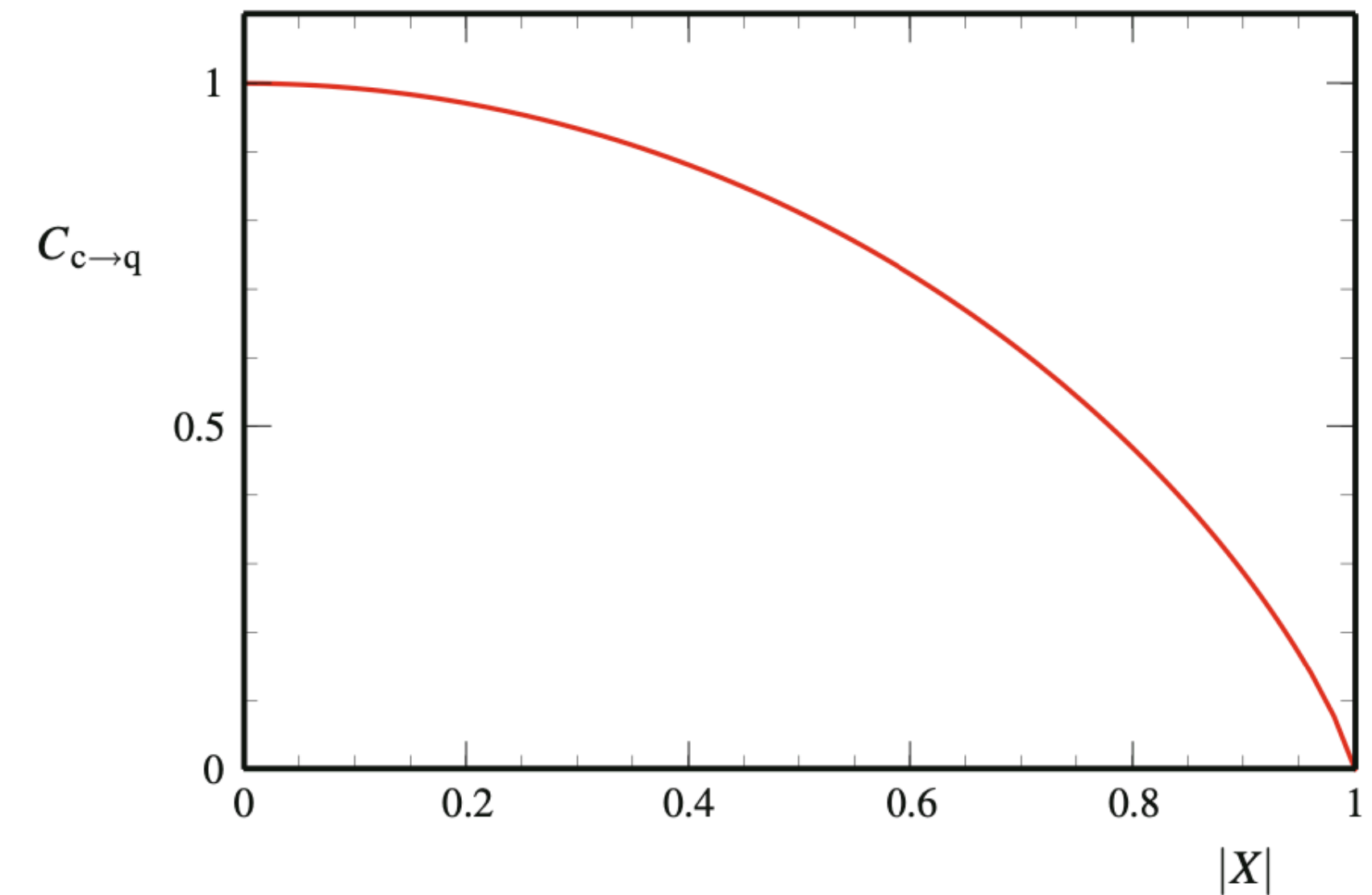
Maximum is obtained when the states are orthogonal, i.e., for $\theta = \pi/2$

Example: Channel-Quantum Capacity



- **Example:** Pure states qubit system: $\mathcal{L} = \{p_a, |\psi_a\rangle\langle\psi_a|, a = 0,1\}$
- Then Holevo information: $\chi(\mathcal{L}) = S(\rho)$,
- Let $|\psi_0\rangle = \cos \phi |0\rangle + \sin \phi |1\rangle$, $|\psi_1\rangle = \cos \beta |0\rangle + \sin \beta |1\rangle$
- Density Operators:
- Due to symmetry, the optimum is obtained with $p_0 = p_1 = 1/2$.

- Avg. Density operator: $\rho = \begin{bmatrix} \frac{1}{2} [\cos^2 \beta + \cos^2 \phi] & \frac{1}{4}(\sin 2\beta + \sin 2\phi) \\ \frac{1}{4}(\sin 2\beta + \sin 2\phi) & \frac{1}{2} [\sin^2 \beta + \sin^2 \phi] \end{bmatrix}$



Capacity $C_{c \rightarrow q}$ of an input qubit source as a function of the superposition parameter $|X|$ of the input states

- Then $\lambda_{\pm} = \frac{1}{2} \left[1 \pm \sqrt{\cos^2(\beta - \phi)} \right] = \frac{1}{2} [1 \pm |X|]$ where $X = \cos(\beta - \phi)$.
- Thus $C_{c \rightarrow q} = h_2(1/2[1 - |X|])$

References

1. R. Wolf, *Quantum Key Distribution: An Introduction with Exercises*, Lecture Notes in Physics, vol. 988. Switzerland: Springer, 2021.
2. G. Cariolaro, *Quantum Communications*. Switzerland: Springer, 2015.



Thank You